

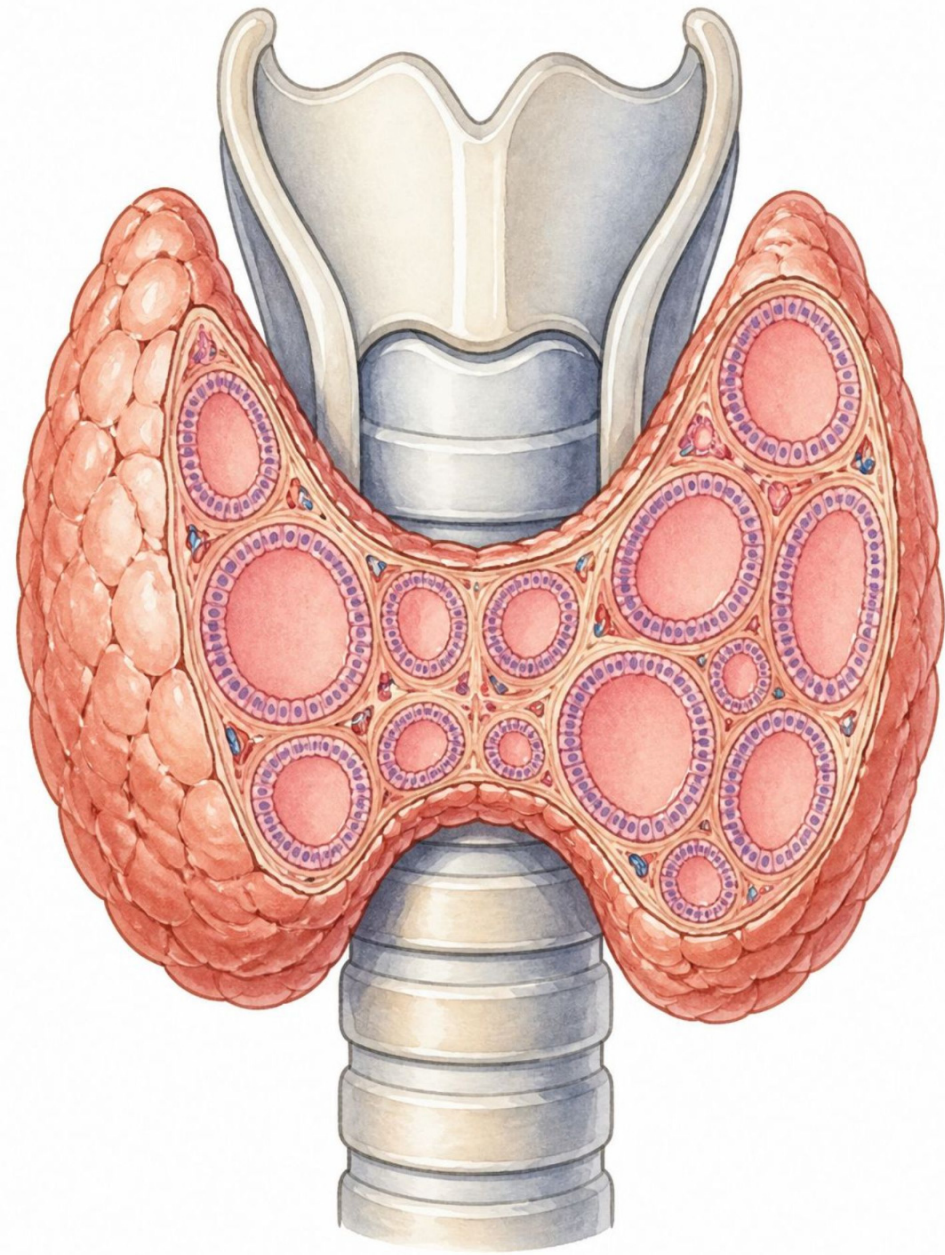


Ultrasound Imaging and TIRADS Classification in Thyroid Nodules

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Outline

- Overview
- What is US imaging ?
- Why US ?
- Anatomy Thyroid
- TIRADS



Overview

Common Finding

Thyroid nodules are frequently encountered during routine ultrasonography in healthy adults.

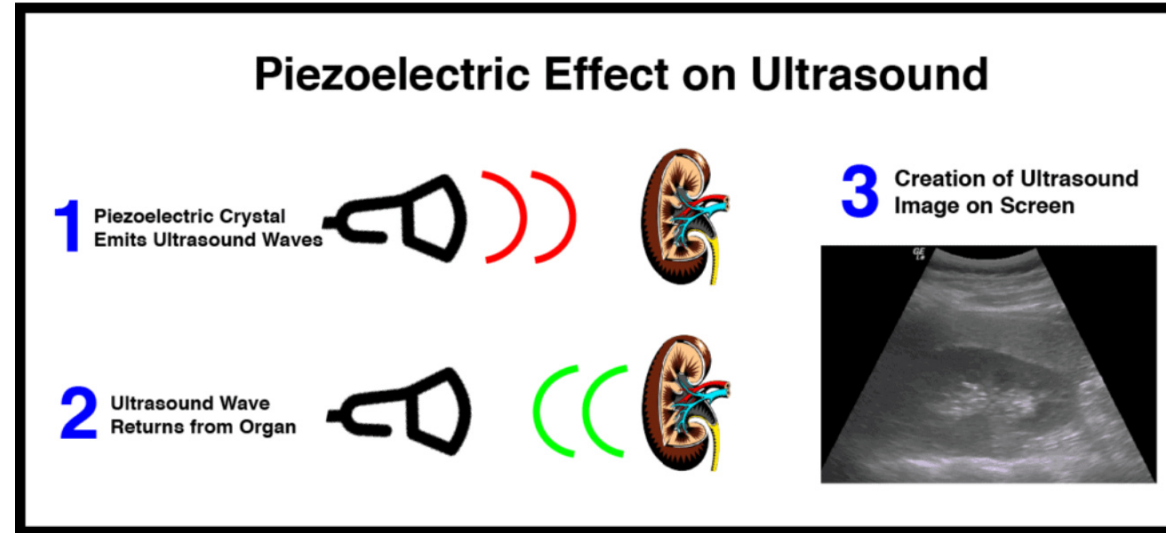
Malignancy Risk

Majority are benign, but carry a **7–15% risk of malignancy** requiring careful evaluation.

What is general US imaging?

- US imaging ☐ sonography ☐ involves exposing part of the body to **high frequency** sound waves ☐ produce images
- US examination do not use ionizing radiation (as used in x-rays).
- US images are captured in **real-time** ☐ show the structure and movement of the body's internal organs ☐ as well as blood flowing through blood vessels

Principles of US



- Optimum frequency range for diagnostic ? 1-20 MHz.
- 3,5-5 MHz: abdominal, heart
- 5-10 MHz: **thyroid**, carotid, breast, testis, and other superficial tissues, infants.
- 10-15 MHz: for the eye

Ultrasound parts

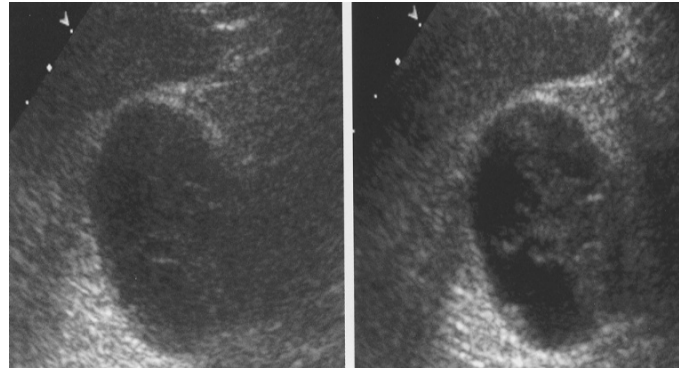


Transducer



Image Optimization

1. Depth
2. Focus
3. TGC (to obtain a smooth grey-scale picture)
4. Zoom / magnification
5. Gain
6. Harmonic imaging ? High contrast, high spatial resolution and low noise



Benefits

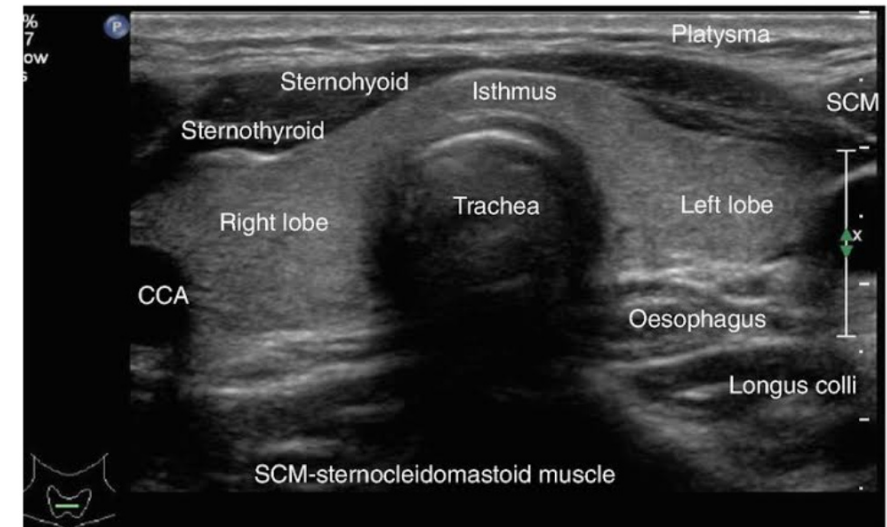
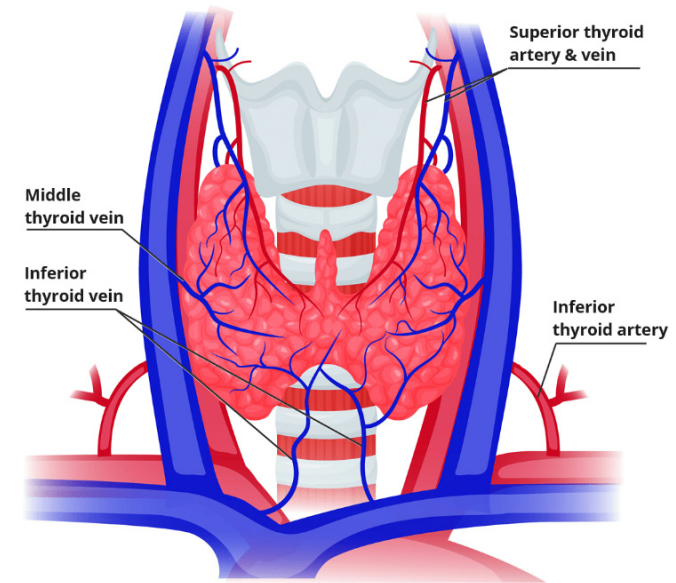
- Mostly noninvasive
- Widely available, easy to use and less expensive (relative)
- Does not use ionizing radiation
- Gives clear soft tissues image better than x ray
- Real-time imaging ☐ guide biopsies

Limitations

- **Operator dependent**
- US waves are disrupted by air / gas ☐ not ideal imaging technique for bowel / organs obscured by the bowel
- Large patients ☐ Low image quality ec greater amounts of tissue attenuates (weakens) the sound waves
- Difficulty penetrating bone

Anatomy Thyroid gland

- Endocrine organ ☐ thyroid hormone production.
- Enveloped by the pretracheal fascia
- Extend from C5 to T1 ☐ anterior to thyroid and cricoid cartilages of larynx
- Butterfly or "H"-shaped ☐ two lobes (each +/- 4 cm in length) ☐ connected by Isthmus.
- Arterial supply
 - A. Superior thyroid (from External carotid)
 - A. Inferior thyroid (from Thyrocervical trunk)
- Venous drainage
 - Superior and Middle thyroid (to Internal jugular vein)
 - Inferior thyroid (to brachiocephalic vein)



ACR TI-RADS: Basic Principles

Goal: Improve the consistency of thyroid nodule evaluation and classification on ultrasound.

Provides specific management recommendations for each thyroid nodule.

Nodules are assessed $\square$ 5 sonographic feature categories:

- Composition
- Echogenicity
- Shape
- Margin
- Echogenic foci

Each feature is assigned 0–3 points according to its association with malignancy risk.

Total score $\rightarrow$ TR category (TR1–TR5) $\rightarrow$ management recommendation (follow-up or FNA).

Why We Need Risk Stratification

The Scope of the Problem

Thyroid nodules are found in as many as **68% of high-resolution ultrasound scans** — making them one of the most common incidental findings in radiology practice.

Without a consistent framework, management decisions become arbitrary and patients face unnecessary procedures or inadequate follow-up.

A Fragmented Landscape

Over the past two decades, professional organizations worldwide developed a **multitude of independent risk stratification systems (RSSs)** — each yielding different management recommendations for identical nodules.

The result: confusion for practitioners and patients alike, and wide variation in biopsy rates across institutions.

TABLE 1: Comparison of ACR TI-RADS, ATA Guidelines, EU-TIRADS, and K-TIRADS

ACR TI-RADS Level	ATA	FNA Threshold (cm)	EU-TIRADS	FNA Threshold (cm)	K-TIRADS	FNA Threshold (cm)
High (TR5)	Solid hypoechoic nodule with any of five suspicious features ^a ; solid hypoechoic component of partial cystic nodule with any suspicious features (both TR4–TR5)	≥ 1.0	At least one of four suspicious features ^b (TR4–TR5)	≥ 1.0	Solid hypoechoic nodule with any of three suspicious features ^c (TR4–TR5)	≥ 1.0
Intermediate (TR4)	Solid hypoechoic nodule with no suspicious features (TR4)	≥ 1.0	No suspicious features, but mildly hypoechoic (TR4)	≥ 1.5	Solid hypoechoic nodule with no suspicious features (TR4); partially cystic or isohyperechoic solid nodule with any of three suspicious features ^c (TR4–TR5)	≥ 1.0
Low (TR3)	Solid isoechoic or hyperechoic nodule with no suspicious features, only macrocalcifications (TR3–TR4); partial cystic nodule with eccentric solid areas with no suspicious features, only macrocalcifications (TR3); partial cystic nodule with eccentric solid areas with no suspicious features or echogenic foci (TR2)	≥ 1.5	No suspicious features, iso- or hyperechoic (TR2–TR3)	≥ 2.0	Partially cystic or isohyperechoic solid nodule with no suspicious features (TR2–TR3)	≥ 1.5
Very low (TR2)	Partially cystic nodule without high, intermediate or low suspicion features, or spongiform (TR1–TR2)	≥ 2.0	Anechoic or spongiform (TR1)	No FNA	Spongiform, partially cystic nodule with comet tail, pure cyst (TR1)	≥ 2.0 spongiform
Benign or normal (TR1)	Purely cystic (TR1)	No FNA	No nodule		No nodule	

Note—ACR TI-RADS level or levels given in parentheses represent the TR level or levels given the description provided by ATA, EU-TIRADS, and K-TIRADS. ACR = American College of Radiology, TI-RADS = Thyroid Imaging Reporting and Data System, ATA = American Thyroid Association, EU-TIRADS = European Thyroid Association Thyroid Imaging Reporting and Data System, K-TIRADS = Korean Society of Thyroid Radiology Thyroid Imaging Reporting and Data System, FNA = fine-needle aspiration.

^aThe five suspicious features are irregular margin, microcalcifications, taller-than-wide shape, rim calcifications with extrusive soft tissue, and extrathyroidal extension.

^bThe four suspicious features are irregular shape, irregular margin, microcalcifications, and marked hypoechogenicity.

^cThe three suspicious features are irregular or spiculated margin, microcalcifications, and taller-than-wide shape.

ACR TI-RADS: The Points-Based Approach

Why Points-Based?

Ensures every nodule receives a classification.

By contrast, ATA guidelines left **13.9% of nodules unclassifiable**, while K-TIRADS left 3.9% unclassified. ACR TI-RADS eliminates this gap entirely.

Five risk tiers (TR1–TR5) map directly to clear FNA and surveillance thresholds, reducing ambiguity at the point of care.

Key Advantage: Higher Size Thresholds

The Evidence Base

Thyroid cancer carries a **10-year relative survival of 99.4%** for tumors smaller than 3 cm.

Mortality does not exceed baseline until tumors surpass 2.5 cm.

Larger Biopsy Cutoffs

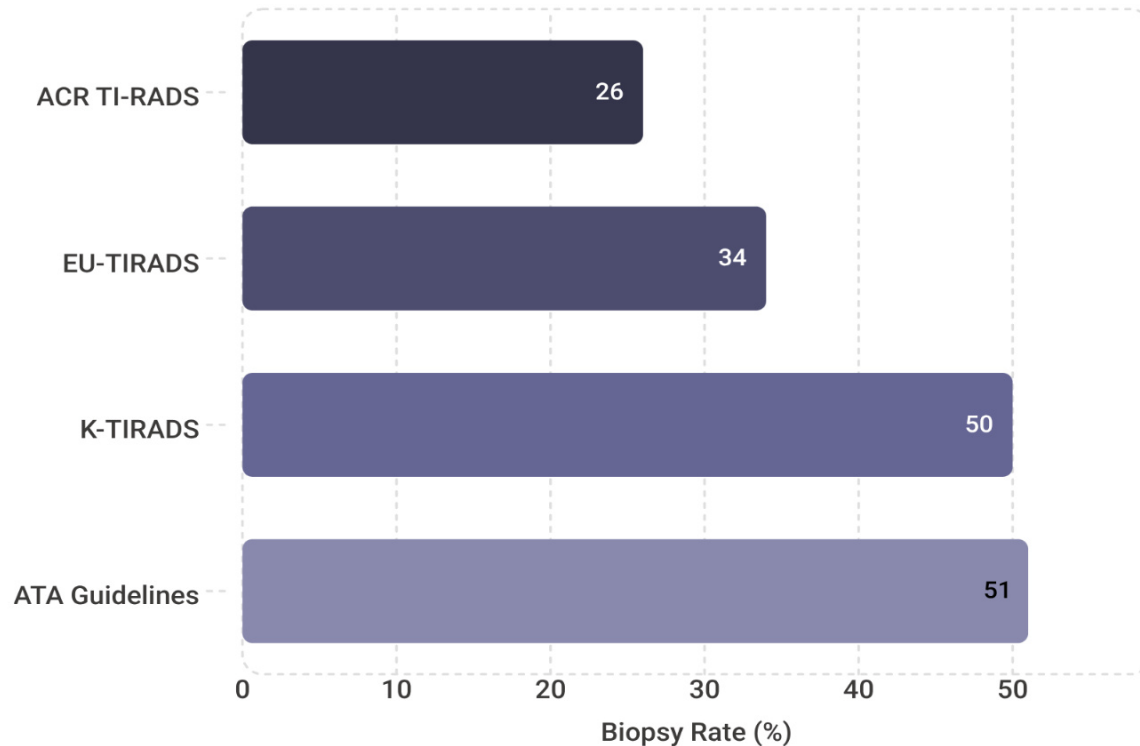
ACR TI-RADS sets the TR3 FNA threshold at **2.5 cm** — compared to 1.5 cm under ATA guidelines — directly reflecting this survival data.

Biopsy Reduction

Grani et al.: The higher thresholds translate to a **19.9–46.5% reduction** in unnecessary biopsies compared to other leading risk stratification systems, with the **lowest false-negative rate (2.2%)**

Evidence: Biopsy Reduction Across Studies

RSS System



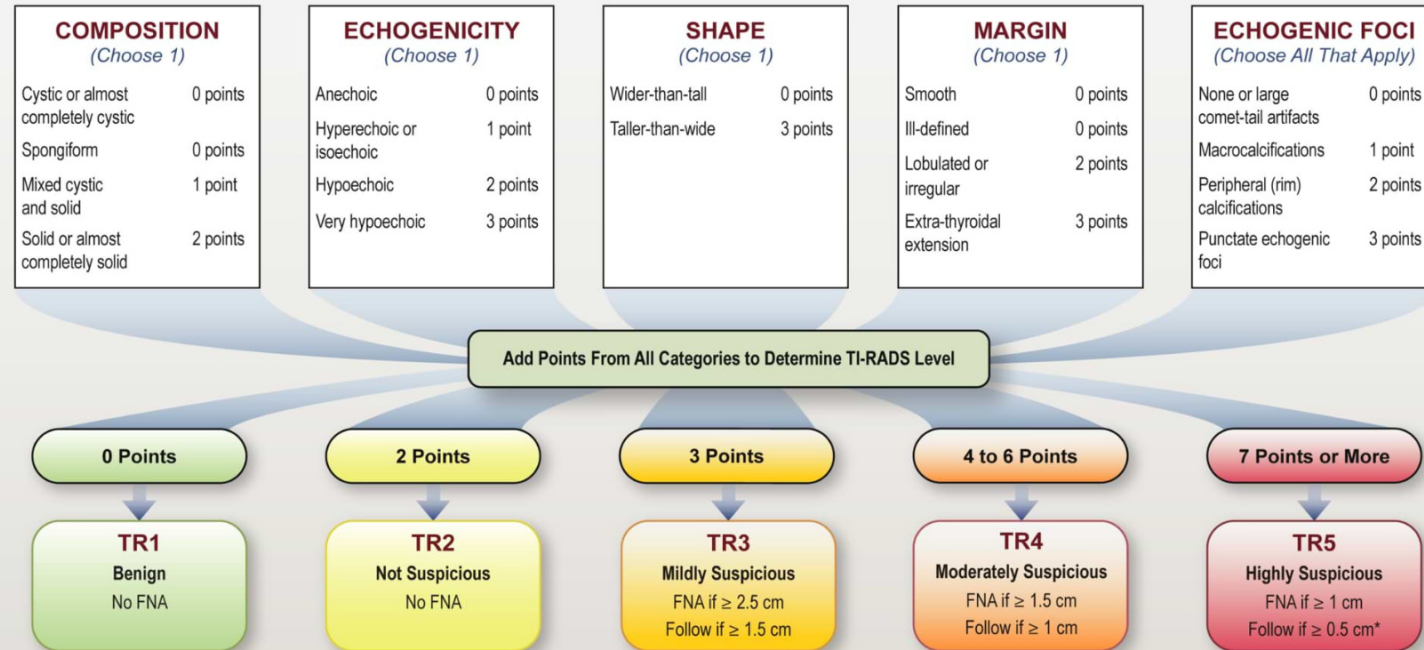
ACR Registry: 32,746 Nodules

In the largest comparative analysis, ACR TI-RADS recommended biopsy in only **26% of nodules** — half the rate of ATA guidelines (51%) and K-TIRADS (50%).

Grani et al. (502 Nodules)

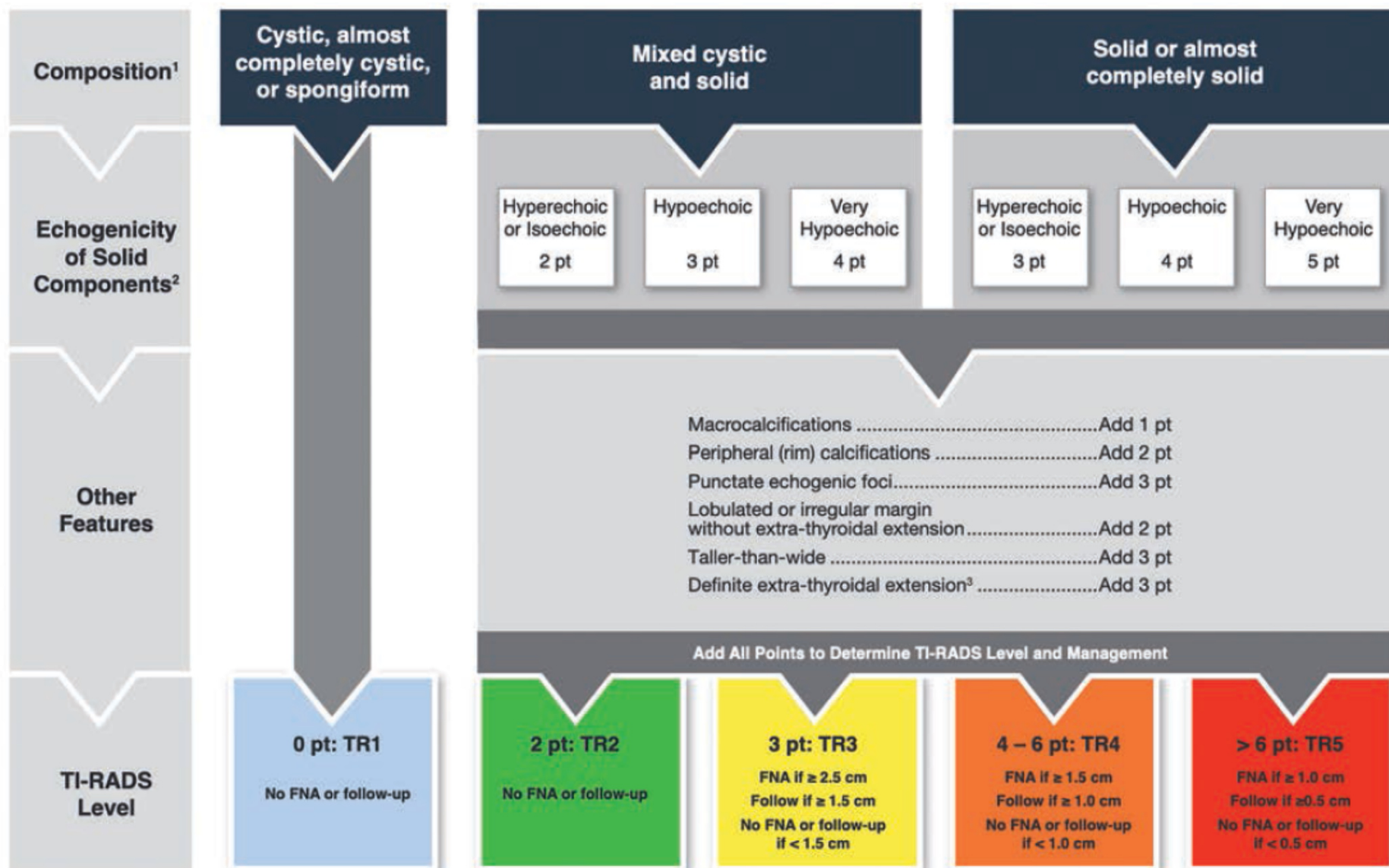
ACR TI-RADS achieved the **lowest benign biopsy rate at 43.8%**, versus 54.7% for ATA, 68.0% for EU-TIRADS, and 82.2% for K-TIRADS — reducing unnecessary procedures by 17.1–53.4%.

ACR TI-RADS



COMPOSITION	ECHOGENICITY	SHAPE	MARGIN	ECHOGENIC FOCI
<i>Spongiform</i>: Composed predominantly (>50%) of small cystic spaces. Do not add further points for other categories. <i>Mixed cystic and solid</i>: Assign points for predominant solid component. Assign 2 points if composition cannot be determined because of calcification.	<i>Anechoic</i>: Applies to cystic or almost completely cystic nodules. <i>Hyperechoic/isoechoic/hypoechoic</i>: Compared to adjacent parenchyma. <i>Very hypoechoic</i>: More hypoechoic than strap muscles. Assign 1 point if echogenicity cannot be determined.	<i>Taller-than-wide</i>: Should be assessed on a transverse image with measurements parallel to sound beam for height and perpendicular to sound beam for width. This can usually be assessed by visual inspection.	<i>Lobulated</i>: Protrusions into adjacent tissue. <i>Irregular</i>: Jagged, spiculated, or sharp angles. <i>Extrathyroidal extension</i>: Obvious invasion = malignancy. Assign 0 points if margin cannot be determined.	<i>Large comet-tail artifacts</i>: V-shaped, >1 mm, in cystic components. <i>Macrocalcifications</i>: Cause acoustic shadowing. <i>Peripheral</i>: Complete or incomplete along margin. <i>Punctate echogenic foci</i>: May have small comet-tail artifacts.

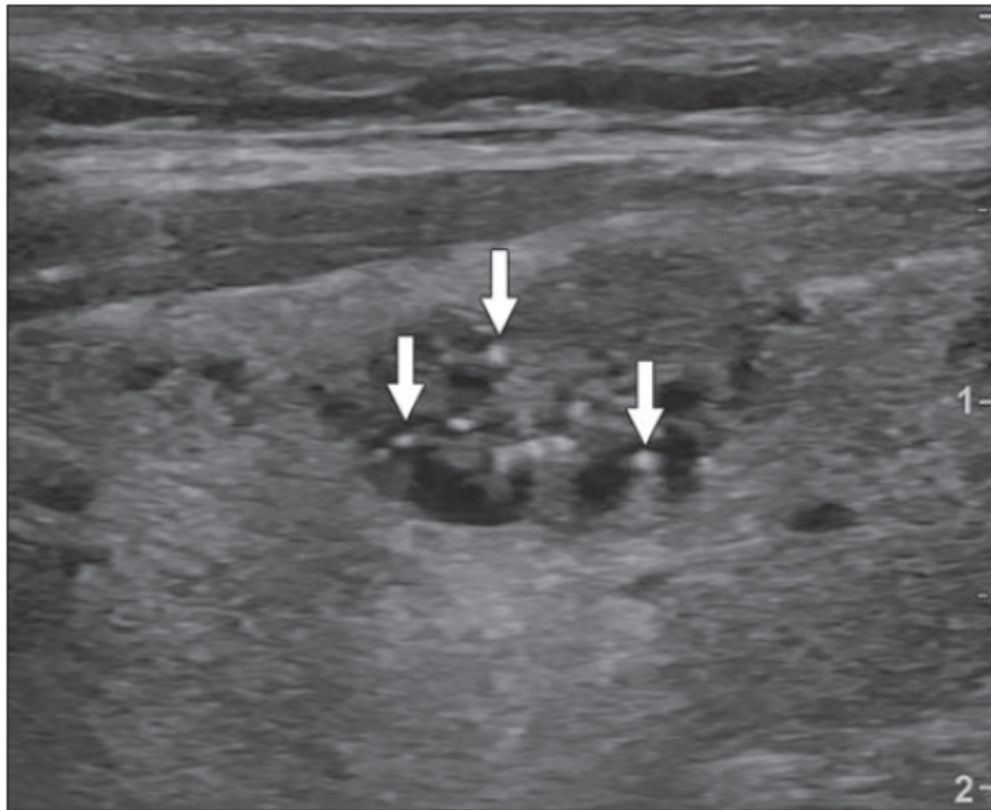
*Refer to discussion of papillary microcarcinomas for 5-9 mm TR5 nodules.



¹ Classify nodule as solid if composition cannot be determined

² Classify nodule as isoechoic if echogenicity cannot be determined

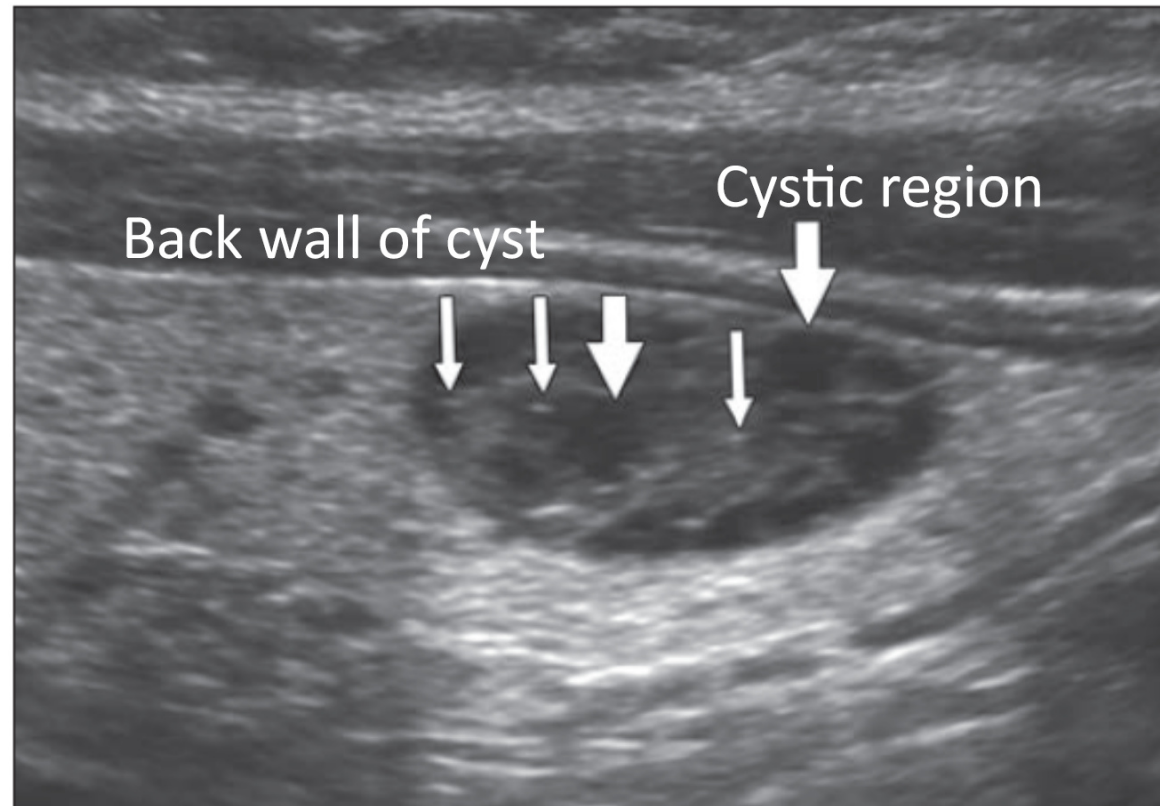
³ Nodules with definite extra-thyroidal extension should be considered malignant until proven otherwise



Mixed cystic-solid (1)
 Isoechoic (1)
 Comet tail artifact (0)

TI-RADS 2

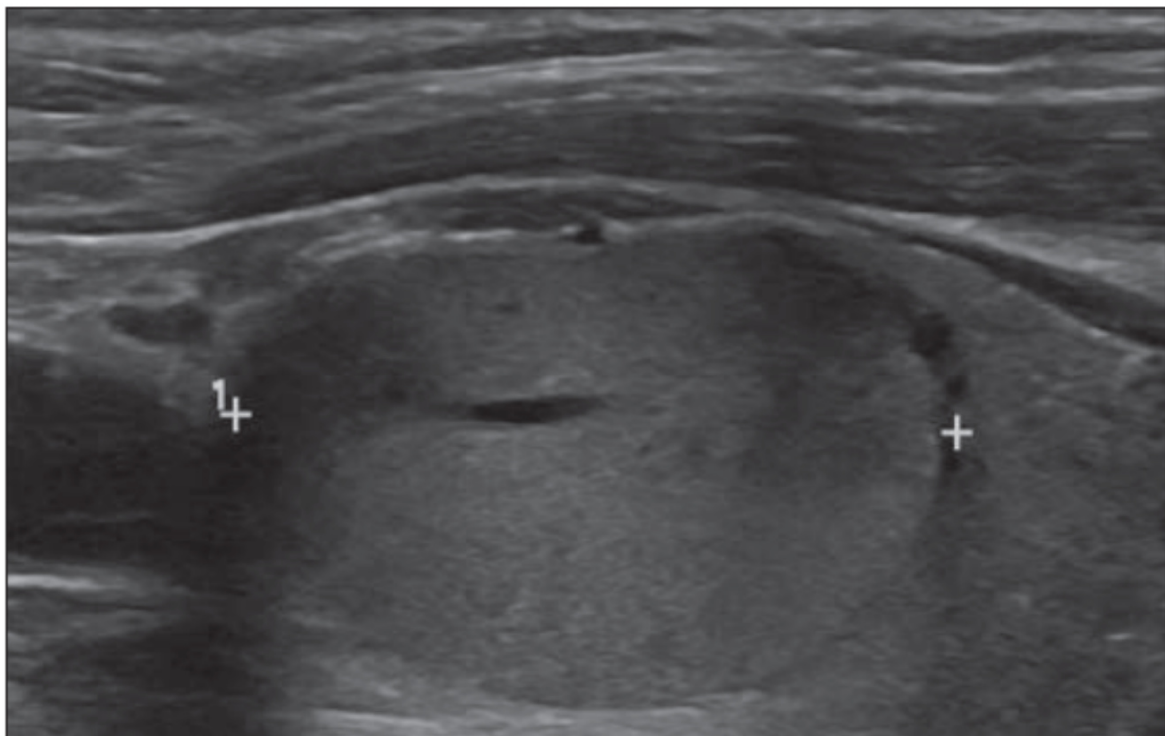
No FNA or follow up



Mixed cystic-solid (1)
 hypoechoic (3)

TI-RADS 3

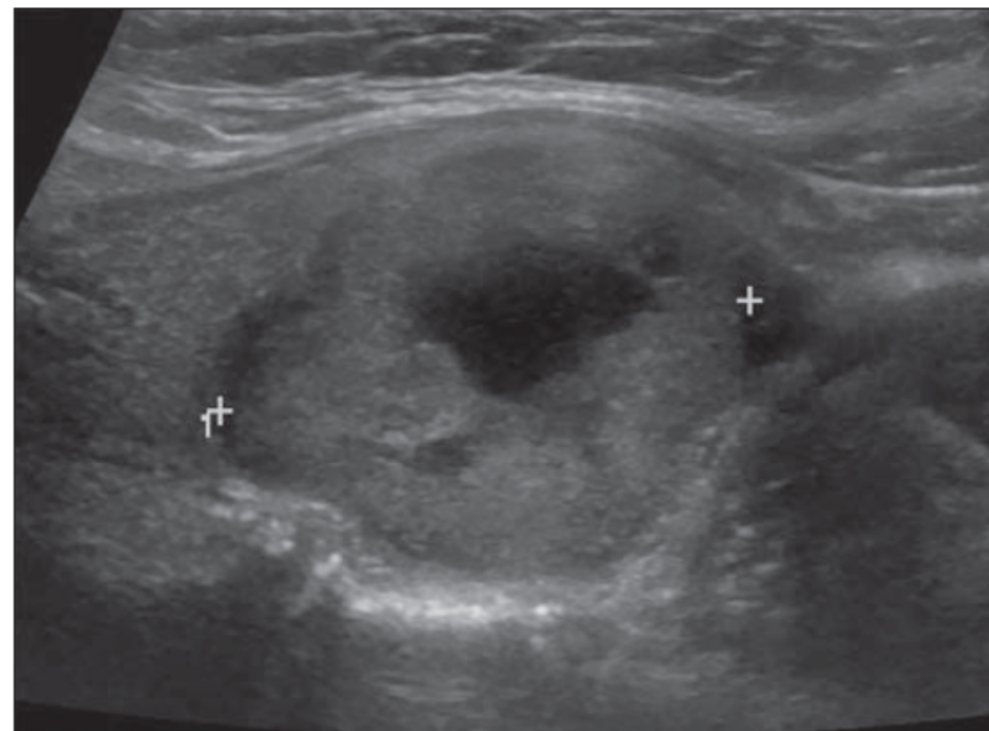
FNA at 2.5 cm and
 follow up at 1.5 cm



Solid (2)
Isoechoic (1)

TI-RADS 3

FNA at 2.5 cm and
follow up at 1.5 cm



Mixed cystic-solid (1)
Isoechoic (1)
Lobulated margin (2)

TI-RADS 4

FNA at 1,5 cm and
follow up at 1.0 cm

Safety: Will We Miss Cancers?

The Core Concern

Reducing biopsies raises the legitimate question: are we deferring diagnosis of clinically significant malignancies?

The Built-In Safety Net

ACR TI-RADS includes **explicit follow-up recommendations** for all nodules that do not meet FNA thresholds. Surveillance is a proven safe strategy for small suspicious nodules and even diagnosed small papillary thyroid cancers, given their indolent biology.

The Numbers

68.2% of malignancies received biopsy recommendations. An additional **21% were captured through surveillance monitoring** — yielding a combined detection pathway of **89.2%** of malignancies addressed under the protocol.

The Interobserver Variability Challenge

Structured Reporting Is Not Enough

Despite the explicit points-based framework, a study of **100 nodules interpreted by eight radiologists** revealed only moderate agreement for management decisions ($\kappa = 0.51$).

Biopsy rates across **seven different practices** using ACR TI-RADS ranged from **4% to 35%** — a nine-fold spread that underscores how much interpretation varies in real-world settings.

The Weakest Links

Margin

$\kappa = 0.25\text{--}0.39$

Fair agreement only

Echogenic Foci

$\kappa = 0.25\text{--}0.39$

Most consequential feature

Management

$\kappa = 0.51$

Moderate overall

Sources of Variability: Artifacts & Technique

Comet Tail Artifacts

Large comet tail artifacts — or high-echogenicity foci from the back wall of tiny cysts — are frequently **misread as punctate echogenic foci (PEF)**. Since PEF carry 3 points, this single misclassification can elevate a TR2 nodule to **TR4**, triggering biopsy.

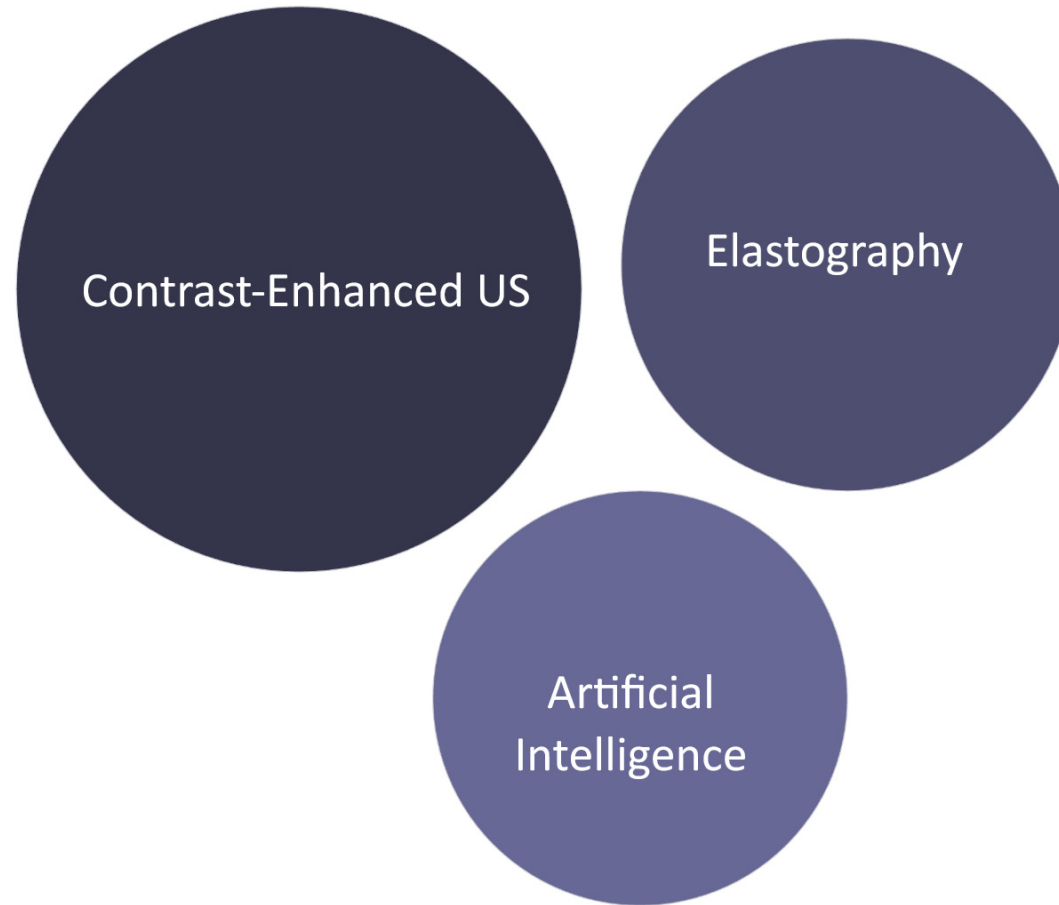
Scanning Parameters

Gain settings, transducer frequency, and post-processing compression **significantly affect feature conspicuity**. The same nodule can appear differently depending on equipment and operator technique — before interpretation even begins.

Training & Experience

Reader expertise matters. **Training sessions and consensus discussions** demonstrably improve feature assignment accuracy and reduce inter-reader discordance across institutions.

Future Directions: Emerging Technologies



Each technology addresses a distinct limitation of conventional B-mode ultrasound. Elastography offers reproducible stiffness quantification (sensitivity 75–89% in meta-analyses), contrast-enhanced ultrasound interrogates tumor vascularity, and AI-driven point optimization has already demonstrated accuracy gains beyond unmodified ACR TI-RADS — though adoption barriers of time, cost, and regulatory approval remain.

Thank You

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